

SECTION 1: QUICK-RESPONSE SAFETY MATRIX (THE "FIRST LOOK")

IMPORTANT: This section constitutes the mandatory "Safe-Start" zone under the *Electricity (Electric Vehicle Charging Station) Safety Regulations 2026*. Operators and users **must** review these hazards before initiating the automated docking sequence.

1.1 HAZARD CLASSIFICATION

The following symbols identify the primary risks associated with **IEC 61851-26:2026** underbody charging systems.

Icon	Classification	Risk Description
	HIGH VOLTAGE DC	System operates at > 400V DC. Direct contact with exposed components or arcing during docking failure can result in fatal electrocution.
	MECHANICAL PINCH POINTS	The automated robotic docking arm exerts high torque. Keep feet, hands, and foreign objects clear of the underbody perimeter during deployment.
	THERMAL & FIRE RISK	Rapid DC transfer generates significant heat. In the event of a cooling system failure, surfaces may exceed 60°C. High risk of localized fire if insulation is breached.

 1.2 STOP-ACTION PROTOCOLS (EMERGENCY ONLY)

IN THE EVENT OF SMOKE, SPARKS, OR MECHANICAL MALFUNCTION, EXECUTE THE FOLLOWING STEPS IMMEDIATELY.

1.2.1 Immediate Emergency Sequence



1. **DEPRESS THE EPO BUTTON:** This physically severs the contactor circuit independently of the software.
2. **DRIVER OCCUPANT SAFETY:**
 - **If Inside:** Stay in the vehicle. The car's tires and chassis act as a Faraday cage. Do not step out until the ground is verified safe.
 - **If Outside:** Move at least 5 meters away immediately.
3. **VERIFY DE-ENERGIZATION:** Do not rely on LEDs. Check the **Mechanical Gravity-Drop Flag** on the pylon side-panel. If the flag is **DOWN**, the physical contactors have dropped.

1.2.2 Manual Override & Extraction (Mechanical Only)

- **Driver Extraction:** If the vehicle is immobilized by a stuck arm, the operator must engage the **Physical Isolation Lever** first.
- **Mechanical Retraction:** Use the hex-key tool to rotate the gear. This system is geared **100:1** to ensure manual force can overcome robotic motor resistance without electrical aid.
- **Visual Confirmation:** Ensure the arm is physically seated in the recessed housing before the driver exits the vehicle.

1.2.2 Manual Override (Underbody Arm Retraction)

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1.3 TRILINGUAL QUICK-WARNINGS

English	Shona	IsiNdebele
DANGER: High Voltage. Keep Clear.	NGOZI: Magetsi ane simba. Pambuka.	INGOZI: Qapela ugetsu.
STOP: Press button in emergency.	MIRA: Dzvanya bhatani kana paita njodzi.	MAMELA: Uma kunesimo esiphuthumayo, cindezela inkibho.

SECTION 2: TECHNICAL OPERATIONAL ENVIRONMENT

2.1 PRE-CONDITIONING REQUIREMENTS (VEHICLE STATUS)

The Vehicle Control Unit (VCU) must broadcast the following statuses via **ISO 15118-20** before the docking arm is energized:

- b ☒

Electronic Parking Brake (EPB) must be fully engaged.

Transmission Lock: Drive selector must be in the **"PARK" (P)** position.

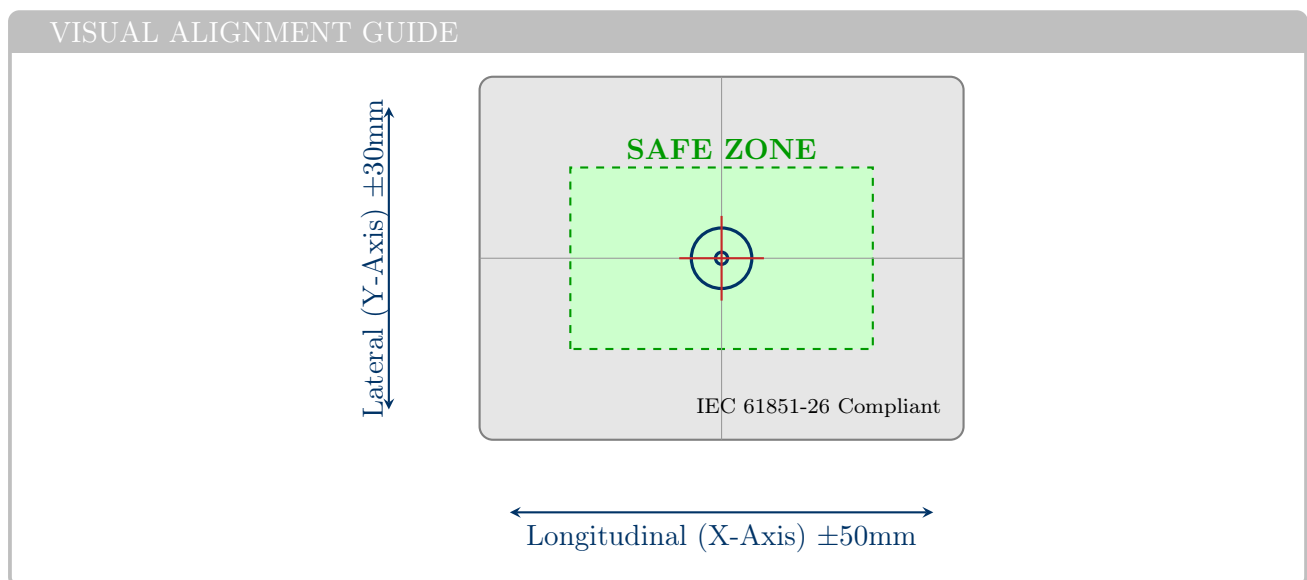
Ignition Status: Propulsion system must be switched **"OFF."**

Charging Port Access: Underbody shield must be in the **"Open"** or **"Retracted"** state.

2.2 ALIGNMENT TOLERANCE (X/Y AXIS VARIANCE)

Safe docking depends on the precise alignment of the vehicle's secondary coil with the station's docking head.

Axis	Permissible Variance	Corrective Action
Longitudinal (X-Axis)	± 50 mm	Driver must roll vehicle forward/backward.
Lateral (Y-Axis)	± 30 mm	Driver must re-center vehicle within the bay lines.
Angular (Yaw)	$< 3^\circ$	Ensure vehicle is not parked at an oblique angle.



2.3 ENVIRONMENTAL CONSTRAINTS (ZIMBABWEAN CORRIDOR)

Specific climatic inhibitors and contingency protocols for regional corridors:

≡ **2.3.1 Flash Flooding:** Docking prohibited if standing water on pad exceeds **30mm**. Ultrasonic sensors will inhibit arm deployment. **System Action:** Station will broadcast "Submerged" status to the national map to reroute traffic.

 **2.3.2 Heavy Dust (Sensor Integrity):** If optical alignment sensors are obscured by dust (common August–October):

- **Safety Protocol:** **DO NOT** attempt manual cleaning. The robotic arm is high-torque and dangerous.
- **Auto-Recovery:** The system will initiate an automated compressed-air "Lens Purge" sequence.
- **Failure State:** If cleaning fails, the bay will lock in "Out of Service" mode and alert a certified technician.

 **2.3.3 Thermal Throttling & Traffic Contingency:** If ground temperatures exceed **50°C**, charging is throttled to **50% capacity**.

- **Congestion Protocol:** The Station Controller will activate the **Peak Heat Rerouting handshake**.
- **User Action:** Incoming vehicles will receive a mobile notification to proceed to the nearest "Cool Zone" station or shaded bay to minimize corridor congestion.

2.4 DIGITAL FENCE INTERLOCK STATUS

Indicator Light	System State	Meaning
 Solid Blue	STANDBY	Waiting for vehicle to enter the Digital Fence.
 Flashing Blue	ALIGNING	Robotic arm is calculating X/Y variance.
 Solid Green	ENGAGED	Safe connection established; HV flow active.
 Flashing Red	BLOCK	Charging inhibited due to weather or alignment error.

SECTION 3: AUTOMATED DOCKING SEQUENCE

THE "PHASE-GATE" APPROACH

The system utilizes a linear safety protocol. Progression to the next "Gate" is prohibited unless specific parameters of **IEC 61851-26:2026** are verified by the Vehicle Control Unit (VCU).

3.1 SEQUENCE OVERVIEW

Phase	Mechanical/Digital Action	Safety Check (IEC 61851-26)
01: Discovery	Arm exits housing; moves toward underbody.	Obstacle Detection: LIDAR active. Motion stops if object detected.
02: Engagement	Connector makes contact; initiates lock.	PE Verification: Protective Earth continuity confirmed.
03: Insulation	Low-current pre-charge pulse sent.	R_{iso} Check: Verifies isolation resistance ($> 500\Omega/V$).
04: Energize	High-voltage DC flow commences.	Thermal Gate: Session kills if connector exceeds 90°C .

3.2 DETAILED PHASE DESCRIPTIONS

PHASE 01: DISCOVERY & PROXIMITY

The station initiates a low-power wireless "ping" to the vehicle. 3D mapping locates the intake port.

User Action: Remain stationary. Do not open doors or adjust vehicle suspension height.

PHASE 02: MECHANICAL COUPLING (THE "CLICK")

Arm extends pins into the vehicle socket. A mechanical "click" signifies the locking pin engagement.

Safety Gate: If the "Lock-Post" sensor fails within 5s, the arm retracts for a second attempt.

PHASE 03: ELECTRICAL INTEGRITY TEST

The station performs a "leakage test" to ensure the cables and battery are properly sealed and no power is "leaking" into the car's frame.

Safety Gate: Ensures user protection from shock before contactors close.

PHASE 04: FULL POWER ENERGIZATION

System ramps to requested kW. The indicator pylon will pulse **GREEN**.

Safety Gate: "Heartbeat" communication every 20ms. If lost, power is cut in $< 100\text{ms}$.

3.3 SESSION TERMINATION & RETRACTION

EXIT PROTOCOL

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1. **Current Ramp-Down:** Power reduces to 0kW.
2. **Contactors Open:** HV circuits physically disconnected.
3. **Unlock:** Mechanical locking pin retracts.
4. **Home Return:** Arm returns to weather-sealed housing.

❗ DO NOT DRIVE AWAY UNTIL THE "SAFE TO DEPART" LIGHT TURNS GREEN.

DOCUMENT REF: 2026-SEQ-003 | PHASE-GATE OPERATIONAL STANDARDS

SECTION 4: TRILINGUAL SAFETY MESSAGING

In accordance with the **ZERA Inclusivity Mandate (2026)**, all automated infrastructure must display safety instructions in Zimbabwe's primary languages. This ensures accessibility across all Alternative Fuel Corridors (AFCs).

4.1 CRITICAL OPERATIONAL WARNINGS

⚠ WARNING 1: PROXIMITY SAFETY

English: Do not exit the vehicle while the docking arm is in motion.

Shona: Musabude mumotokari apo ruoko rwemuchina rwuri kushanda.

IsiNdebele: Lingaphumi emotweni nxa ingalo yomtshina isasebenza.

🔥 WARNING 2: FIRE & ARCING EMERGENCY

English: In case of smoke or sparks, hit the Emergency Stop.

Shona: Kana pachinge paita chiutsi kana pfarivari, dzvanyai bhatani rekumisa muchina nekuchimbidza.

IsiNdebele: Nxa kulomsuthu loba amanyenzezi omlilo, hlikihla ibhatini yokumisa umtshina masinyane.

4.2 QUICK-REFERENCE SAFETY TABLE

For use in printed Quick-Start Guides and interior vehicle dash-cards.

Instruction	English (Bold)	Shona	IsiNdebele
	Park First	Netesai kuti motokari yakapakwa.	Qiniseka ukuthi imota ipakwe kuhle.
	Stay Clear	Garai kure nemugero we-muchina.	Dedani khatshana lomgodi womtshina.
	No Entry	Vanotenderwa chete ndivo vanopinda.	Abavunyelweyo kuphela abangangena.
	Call Help	Fonera anotungamira panzvimbo ino.	Fonela omkhulu walapha.

4.3 VISUAL HIERARCHY COMPLIANCE

Designers Note: This document adheres to **ZERA-approved formatting**:

- **Font Weight:** Primary English text is **Bold**; vernacular translations are standard weight for rapid scanning.
- **Placement:** All messaging must be legible from **2 meters**, ensuring clarity for a driver seated within an EV.
- **Color Coding:** Yellow for Caution; Red for Immediate Emergency.

ZERA COMPLIANCE OFFICE
Alternative Fuel Corridor Standards

STATION OPERATOR
Certified 2026 Regulation

SECTION 5: MAINTENANCE & REGULATORY COMPLIANCE

This section establishes the mandatory "**Paper Trail**" for ZERA inspections under the *Electricity (Electric Vehicle Charging Station) Safety Regulations 2026*. Failure to maintain these records will result in removal from the National AFC network.

5.1 PREDICTIVE MAINTENANCE & DYNAMIC INSPECTION (PdM)

In accordance with 2026 AFC Safety Mandates, this station utilizes usage-based telemetry alongside physical verification.

Maintenance Tier	Metric/Requirement	Frequency
Real-Time Telemetry	Cloud-based monitoring of arm torque, R_{iso} , and thermal drift.	Continuous (24/7)
Usage-Based Cycle	Physical inspection of mechanical seals and gear lubrication.	Every 500 Docks
Mandatory Safety Audit	Comprehensive LIDAR recalibration and structural stress test.	Every 14 Days

5.1.1 OPERATIONAL LOGBOOK (USAGE-BASED)

Technicians must record the "System Odometer" reading (total dockings) during each physical intervention.

Date	Dock Count	Critical Component Status	Tech ID / Cloud Sync
		Torque Calib. Seal Integrity	
		Torque Calib. Seal Integrity	

⚠ MAINTENANCE STANDARD: Any "Fail" result requires the station to be placed in "Out of Service" mode immediately until secondary verification is performed.

5.2 CERTIFICATION & REGULATORY AUTHORIZATION

5.2.1 ZERA COMPLIANCE STAMP



Validation: _____
Expiry: _____
Inspector Code: _____

5.2.2 DIGITAL VERIFICATION



Scan for Live Status
[map.zera.co.zw/corridors/...](https://map.zera.co.zw/corridors/)

5.3 COMPLIANCE DECLARATION

By operating this facility, the Site Supervisor confirms that all automated docking hardware aligns with the **National Energy Governance Framework 2026**.

Legal Notice: Unauthorized modification to the docking arm software or mechanical safety stops will result in immediate revocation of the operating license and potential legal prosecution under the 2026 Safety Act.

 END OF PROPOSAL 